

▼ Traffic Accident Data Analysis 🚦

◆ Dataset used:

📁 **Road Traffic Accident (RTA) Dataset** – a real-world dataset containing accident details such as time, weather conditions, road surface conditions, area name, and accident severity.

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import zipfile

with zipfile.ZipFile("RTA Dataset.csv.zip", 'r') as zip_ref:
    zip_ref.extractall()

df = pd.read_csv("RTA Dataset.csv")
df.head()
```

	Time	Day_of_week	Age_band_of_driver	Sex_of_driver	Educational_level	Vehicle_driver_relation	Driving_experience
0	17:02:00	Monday	18-30	Male	Above high school	Employee	1-2yr
1	17:02:00	Monday	31-50	Male	Junior high school	Employee	Above 10yr
2	17:02:00	Monday	18-30	Male	Junior high school	Employee	1-2yr
3	1:06:00	Sunday	18-30	Male	Junior high school	Employee	5-10yr
4	1:06:00	Sunday	18-30	Male	Junior high school	Employee	2-5yr

5 rows × 32 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12316 entries, 0 to 12315
Data columns (total 32 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Time                                12316 non-null  object
1   Day_of_week                        12316 non-null  object
2   Age_band_of_driver                 12316 non-null  object
3   Sex_of_driver                     12316 non-null  object
4   Educational_level                  11575 non-null  object
5   Vehicle_driver_relation            11737 non-null  object
6   Driving_experience                 11487 non-null  object
7   Type_of_vehicle                   11366 non-null  object
8   Owner_of_vehicle                  11834 non-null  object
9   Service_year_of_vehicle            8388 non-null   object
10  Defect_of_vehicle                  7889 non-null   object
11  Area_accident_occured              12077 non-null  object
12  Lanes_or_Medians                   11931 non-null  object
13  Road_allignment                    12174 non-null  object
14  Types_of_Junction                 11429 non-null  object
15  Road_surface_type                  12144 non-null  object
16  Road_surface_conditions            12316 non-null  object
17  Light_conditions                   12316 non-null  object
18  Weather_conditions                 12316 non-null  object
19  Type_of_collision                  12161 non-null  object
20  Number_of_vehicles_involved         12316 non-null  int64
21  Number_of_casualties               12316 non-null  int64
22  Vehicle_movement                   12008 non-null  object
23  Casualty_class                     12316 non-null  object
24  Sex_of_casualty                    12316 non-null  object
25  Age_band_of_casualty               12316 non-null  object
26  Casualty_severity                  12316 non-null  object
27  Work_of_casualty                   9118 non-null   object
```

```

28 Fitness_of_casualty      9681 non-null  object
29 Pedestrian_movement     12316 non-null object
30 Cause_of_accident        12316 non-null object
31 Accident_severity        12316 non-null object
dtypes: int64(2), object(30)
memory usage: 3.0+ MB

```

```
df.columns
```

```

Index(['Time', 'Day_of_week', 'Age_band_of_driver', 'Sex_of_driver',
      'Educational_level', 'Vehicle_driver_relation', 'Driving_experience',
      'Type_of_vehicle', 'Owner_of_vehicle', 'Service_year_of_vehicle',
      'Defect_of_vehicle', 'Area_accident_occured', 'Lanes_or_Medians',
      'Road_allignment', 'Types_of_Junction', 'Road_surface_type',
      'Road_surface_conditions', 'Light_conditions', 'Weather_conditions',
      'Type_of_collision', 'Number_of_vehicles_involved',
      'Number_of_casualties', 'Vehicle_movement', 'Casualty_class',
      'Sex_of_casualty', 'Age_band_of_casualty', 'Casualty_severity',
      'Work_of_casualty', 'Fitness_of_casualty', 'Pedestrian_movement',
      'Cause_of_accident', 'Accident_severity', 'Hour', 'Time_of_Day'],
      dtype='object')

```

```
df = df.dropna()
```

```

df['Time'] = pd.to_datetime(df['Time'], format='%H:%M', errors='coerce')
df['Hour'] = df['Time'].dt.hour

```

```

def time_of_day(hour):
    if 6 <= hour < 12:
        return 'Morning'
    elif 12 <= hour < 18:
        return 'Afternoon'
    elif 18 <= hour < 22:
        return 'Evening'
    else:
        return 'Night'

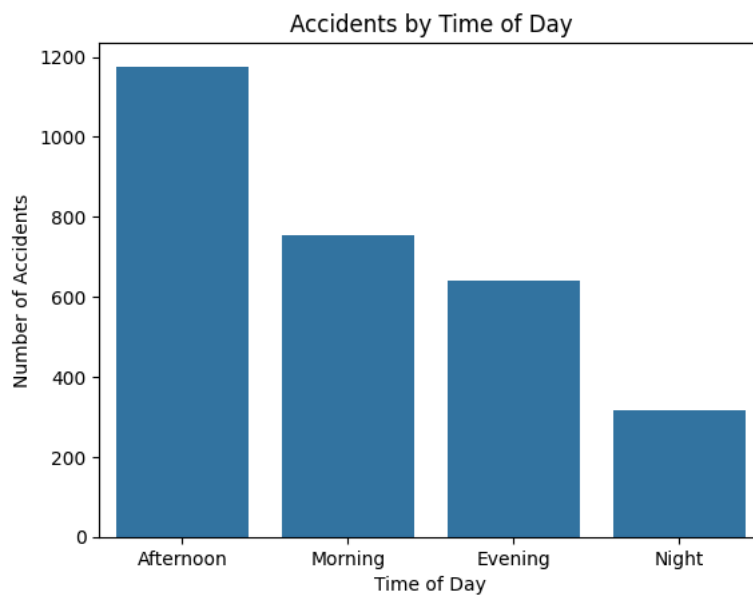
df['Time_of_Day'] = df['Hour'].apply(time_of_day)

```

```

sns.countplot(x='Time_of_Day', data=df)
plt.title("Accidents by Time of Day")
plt.xlabel("Time of Day")
plt.ylabel("Number of Accidents")
plt.show()

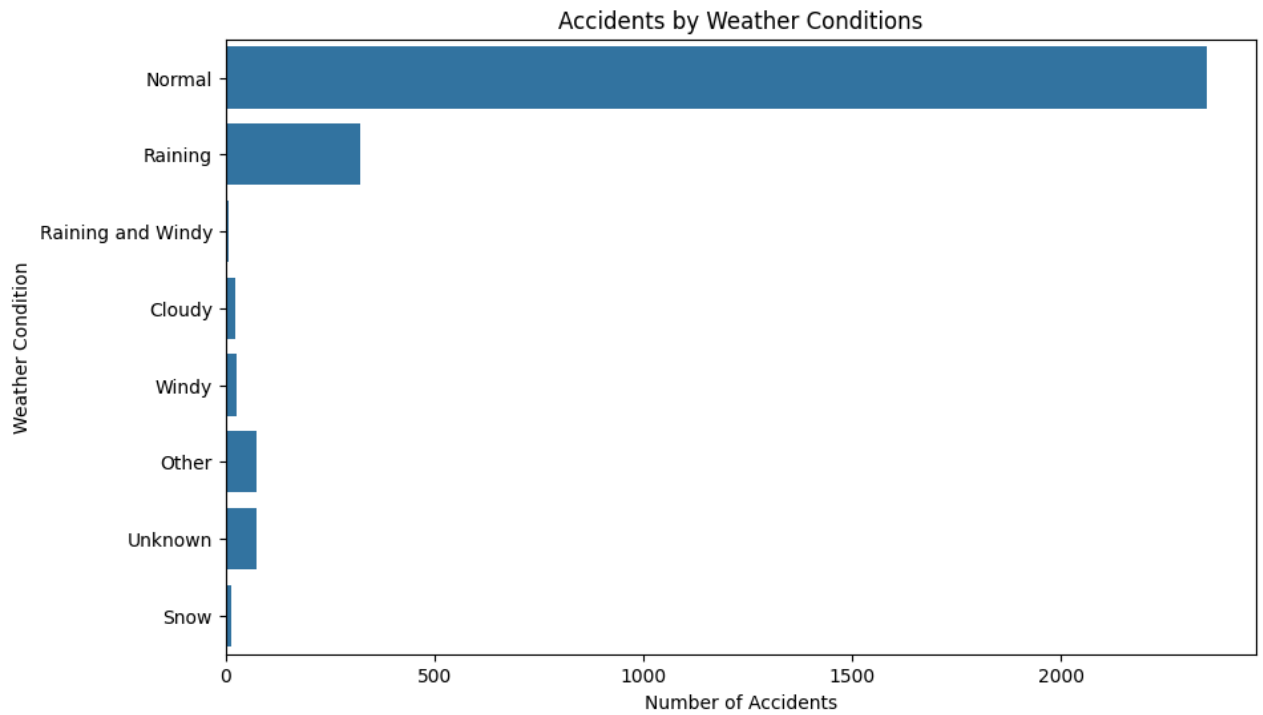
```



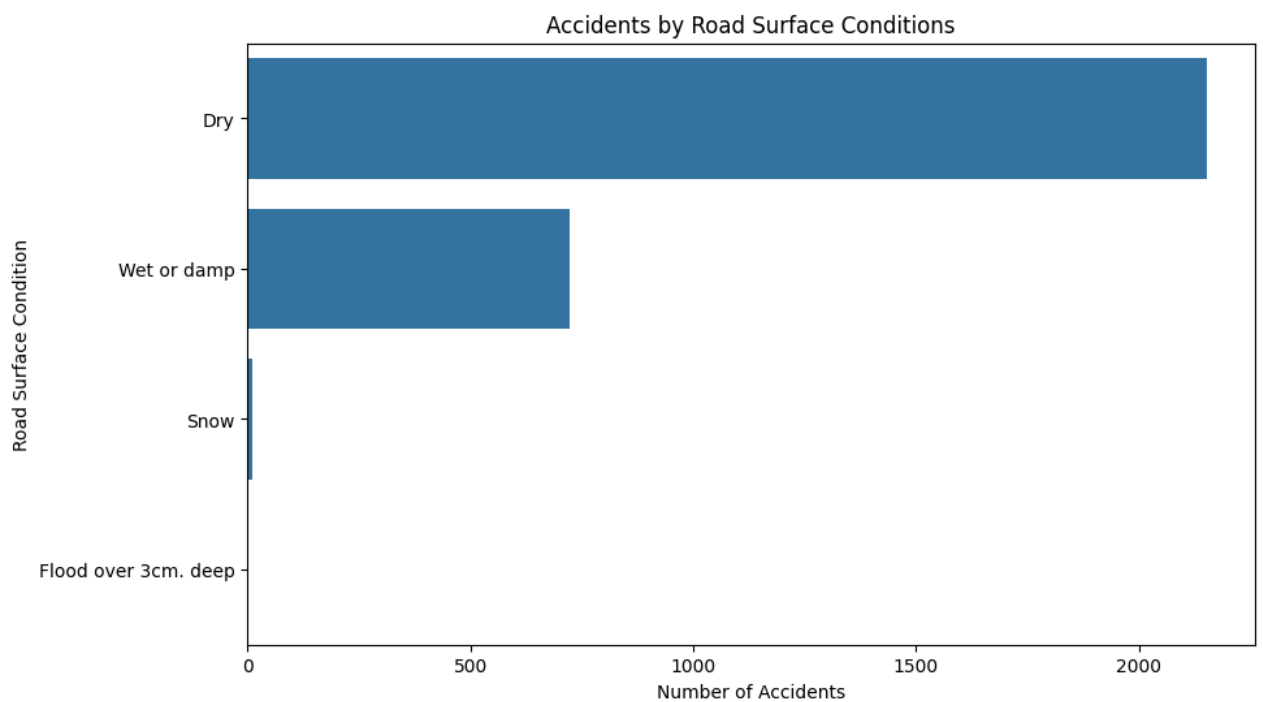
```

plt.figure(figsize=(10,6))
sns.countplot(y='Weather_conditions', data=df)
plt.title("Accidents by Weather Conditions")
plt.xlabel("Number of Accidents")
plt.ylabel("Weather Condition")
plt.show()

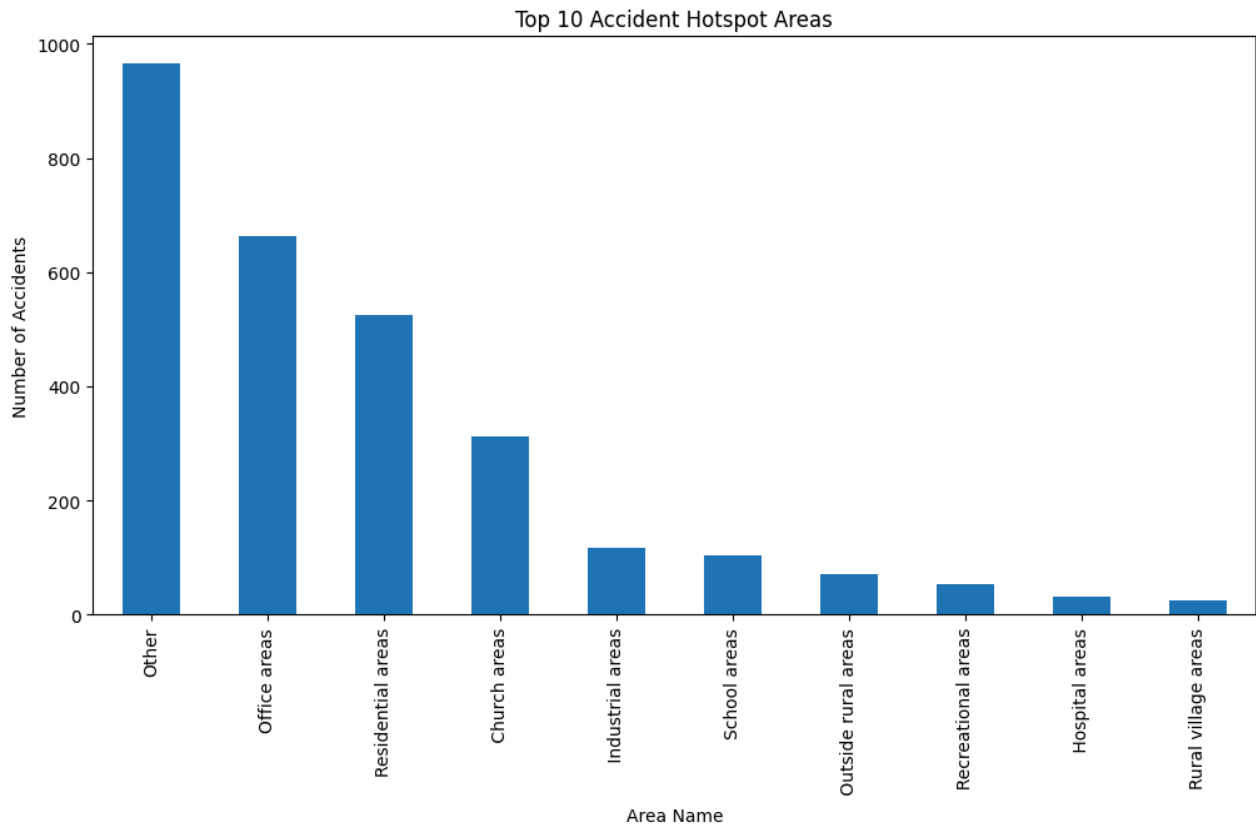
```



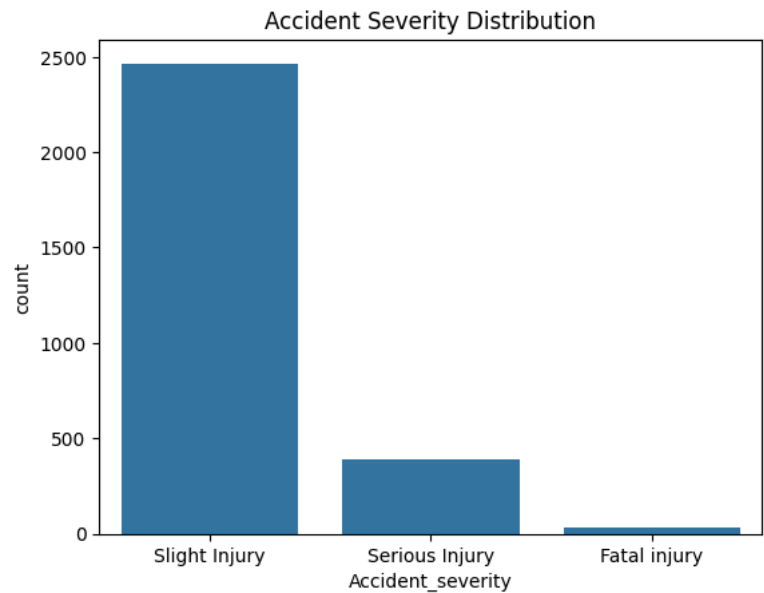
```
plt.figure(figsize=(10,6))
sns.countplot(y='Road_surface_conditions', data=df)
plt.title("Accidents by Road Surface Conditions")
plt.xlabel("Number of Accidents")
plt.ylabel("Road Surface Condition")
plt.show()
```



```
plt.figure(figsize=(12,6))
df['Area_accident_occured'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 Accident Hotspot Areas")
plt.xlabel("Area Name")
plt.ylabel("Number of Accidents")
plt.show()
```



```
sns.countplot(x='Accident_severity', data=df)
plt.title("Accident Severity Distribution")
plt.show()
```



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Final Summary 🚦

- Accident patterns were analyzed based on **time of day 🕒**, **weather conditions 🌧️**, and **road surface conditions 🛣️**

- Most accidents occurred during **morning and evening peak hours**, indicating the impact of traffic congestion

- Weather and poor road conditions were found to increase accident risk

- Area-wise analysis helped identify **accident hotspot locations**

- The study highlights how **data analysis and visualization 📊** can lead to better road safety planning and decision-making

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